

TRANTALOL TABLET 100MG

DESCRIPTION: A yellow, round, 9mm diameter tablet with marking 'duo' and scored on another side.

COMPOSITION: Each tablet contains Labetalol Hydrochloride 100 mg.

PHARMACODYNAMICS:

Labetalol lowers blood pressure by blocking alpha-adrenoceptors in peripheral arterioles and thereby reducing peripheral resistance. Labetalol also concurrently blocks beta-adrenoceptors, notably in the heart, from the reflexly-mediated drive caused by the peripheral vasodilatation. Therefore, the reduction in blood pressure is achieved without cardiac stimulation. Labetalol does not reduce cardiac output at rest, or after moderate exercise. Normally occurring increases in systolic pressure during exercise are, however, lessened by Labetalol corresponding changes in the diastolic pressure are not affected.

PHARMACOKINETICS:

Labetalol antagonises alpha- and beta-adrenoceptors concurrently by competitive inhibition; it has no intrinsic sympathomimetic activity and less pronounced membrane stabilising activity than propranolol. The beta-blockade is non-selective. Labetalol has two centres of chirality which means that four isomers are possible, each of the isomers having different relative alpha- and beta-blocking activities.

Pharmacokinetics: Taken orally, Labetalol tablets are well absorbed. The drug is extensively metabolised by the liver, and possibly in the gut wall, to O-phenyl-glucuronide, N-glucuronide and a glucuronide formed by conjugation at the secondary alcohol group. Peak plasma levels occur at 1-2 hours, associated with a reduction in blood pressure. On average, 35-40% of an administered oral dose reaches the circulation.

The plasma half-life is appropriately 6-8 hours, but a hypotensive effect has been observed up to 11 hours after a given dose.

The absolute bioavailability of labetalol is increased when administered with food.

At therapeutic concentrations, the active drug binds to albumin (52%). Binding also occurs to melanin (0.05%) but the clinical significance is uncertain.

Labetalol and its inactive metabolites are excreted by both the liver and the kidneys, so the drug is unlikely to accumulate in the body.

Whilst impairment of renal function does not require a change in dosage, impairment of liver function requires a reduction in dosage, due to the decrease in first-pass metabolism of the drug.

Labetalol does not adversely affect renal function and is a particularly suitable drug for use in hypertensive patients with renal disease.

INDICATION: Treatment of all grades of hypertension.

RECOMMENDED DOSAGE:

Labetalol tablets are to be taken orally, preferably after meals. For all patients, the dosage of labetalol must be titrated against blood pressure response.

The recommended starting dose for all patients is 100-200 mg twice daily. The dosage should then be increased at weekly intervals until satisfactory control of blood pressure is attained, it is important to follow this dosage instructions and to increase the dosage gradually, in order to minimise side effects. Effective daily doses vary between 200 and 2 400 mg. In those patients on high doses, a regime of 3 to 4 doses a day, may be necessary.

Hypertension may be controlled by labetalol alone, if labetalol tablets are prescribed with diuretics or other antihypertensive agents, an additive effect can be expected.

When transferring patients from other treatment regimes, labetalol should be introduced in the manner described above and the dosage of other treatments should be reduced gradually.

ROUTE OF ADMINISTRATION: Oral

CONTRACINDICATIONS:

Beta-blockers are usually contraindicated in a variety of conditions and as Labetalol has not yet been fully investigated for its effect on these conditions, they should be treated as contraindications until further information is available.

1. Bronchospasm.
2. Allergic disorders (including allergic rhinitis) which may suggest a predisposition to bronchospasm.
3. Right ventricular failure secondary to pulmonary hypertension.
4. Significant right ventricular hypertrophy.
5. Sinus bradycardia (less than 45-50 beats/minute) or sick sinus syndrome.
6. Second- and third-degree A-V block.
7. Cardiogenic shock and hypovolaemic shock.
8. Congestive heart failure.
9. Anaesthesia with agents that produce myocardial depression (e.g., either, chloroform, cyclopropane).
10. Lactation and early pregnancy (see Use in Pregnancy and Lactation).
11. Known sensitivity to labetalol hydrochloride.

WARNINGS & PRECAUTIONS:

Cardiac Failure: Beta blockade depresses myocardial contractility and may precipitate cardiac failure in some patients with a history of cardiac failure, chronic myocardial insufficiency or unsuspected cardiomyopathy. In patients with no history of cardiac failure, continuing depression of the myocardium may lead to cardiac failure. If cardiac failure develops, the beta-blocker should be withdrawn (see Warnings, 2).

(NOTE: Although congestive heart failure has been considered to be a contraindication to the use of beta-blockers, there is a growing literature on the experimental use of beta-adrenergic blocking drugs in heart failure. As further trials are needed to identify which patients are most likely to respond to which drugs, beta-blockers should not normally be prescribed for heart failure outside of specialist centres.)

Abrupt Withdrawal: Care should be taken if beta-blockers have to be discontinued abruptly in patients with coronary artery disease. Severe exacerbation of angina and precipitation of myocardial infarction and ventricular arrhythmias has occurred following abrupt discontinuation of beta-blockade in patients with ischaemic heart disease. Therefore it is recommended that the dosage be reduced gradually over a period of about 8-14 days during which time the patient's progress should be assessed. The drug may be reinstituted temporarily if the angina worsens. If the drug must be withdrawn abruptly, close observation is required. In the peri-operative period, beta-blockers should not be withdrawn, unless indicated.

Concomitant Therapy with Calcium Antagonist: The concomitant use of beta-blockers and calcium antagonists with myocardial depressant and sinus node activity, eg verapamil and to a lesser extent diltiazem, may cause hypotension, bradycardia and asystole. Extreme caution is required if these drugs have to be used together. The dihydropyridine calcium antagonists (eg nifedipine) have a weaker myocardial depressant effect and can be administered cautiously with beta-blockers. If excessive hypotension develops, the calcium antagonist should be stopped or the dosage reduced.

Antiarrhythmic Drugs: Care should be taken when prescribing beta-blockers with antiarrhythmic drugs. Interactions have been reported during concomitant beta-blocker therapy with the Class IA agents disopyramide, and less frequently quinidine; Class IB agents, tocainide, mexiletine and lidocaine; Class IC agents, flecainide and propafenone (not available in Australia); the Class III agent, amiodarone; and the Class IV antiarrhythmic agents.

Beta-blockers: Beta-blockers, even those with apparent cardioselectivity, should not be used in patients with asthma or a history of obstructive disease unless no alternative treatment is available. In such cases, the risk of inducing bronchospasm should be appreciated and appropriate precautions taken. If bronchospasm should occur after the use of Trandate it can be treated with a beta-2-agonist by inhalation, eg salbutamol (the dose of which may need to be greater than the usual dose in asthma), and, if necessary, intravenous

atropine 1 mg. There have been reports of skin rashes and/or dry eyes associated with the use of beta-adrenoceptor blocking drugs. The reported incidence is small and in most cases the symptoms have cleared when treatment was withdrawn. Gradual discontinuance of the drug should be considered if any such reaction is not otherwise explicable.

Peripheral Circulation: Beta-blockade may impair the peripheral circulation and exacerbate the symptoms of peripheral vascular disease.

Hepatic Injury: Severe hepatocellular injury, confirmed by rechallenge in at least one case, occurs rarely with labetalol therapy. The hepatic injury is usually reversible, but hepatic necrosis and death have been reported. Injury has occurred after both short and long term treatment and may be slowly progressive despite minimal symptomatology. Appropriate laboratory testing should be done at the first sign or symptom of liver dysfunction. If there is laboratory evidence of liver injury or the patient is jaundiced, labetalol should be stopped and not restarted.

Prinzmetal Angina: There is a risk of exacerbating coronary artery spasm if patients with Prinzmetal or variant angina are treated with a beta-blocker. If this treatment is essential, it should only be undertaken in a Coronary or Intensive Care Unit.

Euthyroid hyperthyroxinaemia: The effects of beta-blockers on thyroid hormone metabolism may result in elevations of serum free thyroxine (T₄) levels. In the absence of any signs or symptoms of hyperthyroidism, additional investigation is necessary before a diagnosis of thyrotoxicosis can be made.

Effects on the Ability to Drive and Operate Machinery: Patients should be warned that labetalol hydrochloride may cause occasional dizziness or fatigue, therefore resulting in the impairment of ability to drive or operate machinery.

Anaesthesia and the peri-operative period: Beta blockade may have beneficial effects in decreasing the incidence of arrhythmias and myocardial ischaemia during anaesthesia and the post-operative period. It is currently recommended that maintenance beta-blockade be continued peri-operatively. The anaesthetist must be made aware of beta-blockade because of the potential for interactions with other drugs, resulting in severe bradyarrhythmias and hypotension, the decreased reflex ability to compensate for blood loss, hypovolaemia and regional sympathetic blockade, and the increased propensity for vagal-induced bradycardia. Incidents of protracted severe hypotension or difficulty restoring normal cardiac rhythm during anaesthesia have been reported. Modern inhalational anaesthetic agents are generally well tolerated, although older agents (ether, cyclopropane, methoxyflurane, trichlorethylene) were sometimes associated with severe circulatory depression in the presence of beta-blockade.

Labetalol tablets need not be discontinued prior to anaesthesia but patients should receive intravenous atropine prior to induction.

Synergistic effects of labetalol and halothane on cardiac output and blood pressure have been reported.

Diabetes: Beta blockers affect glucose metabolism and may mask some important premonitory signs of acute hypoglycaemia, such as tachycardia.

In patients with insulin or non-insulin dependent diabetes, especially labile diabetes, or with a history of spontaneous hypoglycaemia beta-blockade may result in the loss of diabetic control and delayed recovery from hypoglycaemia. The dose of insulin or oral hypoglycaemic agent may need adjustment.

In one study, there was an increase in mean fasting glucose levels during labetalol treatment but no alteration in insulin activity or response to an oral glucose tolerance test.

Other Metabolic Effects: Beta adrenoceptors are involved in the regulation of lipid as well as carbohydrate metabolism. Some drugs affect the lipid profile adversely although the long-term clinical significance of this change is unknown and the effect appears to be less for drugs with intrinsic sympathomimetic activity.

Use of Catecholamine-Depleting Agents: Concomitant use of drug such as reserpin and guanethidine requires careful monitoring since the added effect of a beta-blocker may produce an excessive reduction on the resting sympathetic nervous tone.

Eye and skin reactions: Various skin rashes and conjunctival xerosis have been reported with beta-blocking agents. Cross reactions may occur between beta-blockers and substitution within the group may not necessarily preclude occurrence of symptoms. During the long term treatment with the beta-blocking drug, practolol, a specific rash bearing a superficial resemblance to psoriasis was occasionally described. In a number of patients affected, this rash was accompanied by adverse effect on the eye (xerophthalmia and / or keratoconjunctivitis) of varying severity. This condition is called the oculomucocutaneous syndrome or practolol syndrome. In a few patients, these eye changes occurred independently of a skin rash. On rare occasions, otitis media, sclerosing peritonitis, pericarditis and pleurisy have been reported.

Although this syndrome has not been observed in patient taking other beta-blockers, the possibility of such side effects occurring should be borne in mind. More recently an occasion between Peyronie's disease are fibrosing induration of the penis and various beta-blockers has been suggested but is not proven.

It has been found in animal studies that labetalol binds to the melanin pigment of the uveal tract. This significance of this in humans is not known but regular checks on visual function should be made as a precaution.

Extensive ophthalmological monitoring of 72 patients treated with labetalol at doses of 300 to 2400 mg daily for between 6 months and 3 years, and routine monitoring of eye complaints from over 6000 patients has not revealed any adverse effect on the eye.

Cross sensitisation: Cross sensitisation may occur between beta-blockers and substitution within the group may not necessarily preclude recurrence of symptoms.

Sclerosing peritonitis: Sclerosing peritonitis has been reported. In association with practolol therapy but not with other beta-blockers or labetalol. Nevertheless, the possibility of such a reaction must be borne in mind.

Allergic manifestations: Allergic manifestations may be exaggerated by beta-blockade (eg. Allergic rhinitis during the pollen season and allergic reactions to bee and wasp stings). Beta-blockers should be avoided if there is a risk of bronchospasm.

Phaeochromocytoma: Labetalol has been shown to be effective in lowering blood pressure and relieving symptoms in patients with phaeochromocytoma. However, paradoxical hypertension responses have been reported in a few patients with this tumour; therefore, use caution when administering labetalol to patients with phaeochromocytoma.

In patients with this condition, an alpha-blocking drug (eg. Phentolamine / phenoxybenzamine) should be considered before administration of labetalol, bearing in mind that this drug has alpha-blocking and beta-blocking properties.

Hyperthyroidism: Since beta-blockade may mask the clinical signs of developing or continuing hyperthyroidism, resulting in symptomatic improvement without any change in thyroid status, special care should be exercised in those patients who are hyper thyroid and are also receiving labetalol as an antihypertensive agent.

Postural hypotension: Severe postural hypotension has occurred in some patients (see Adverse Reactions). Furthermore this may be enhanced by the concurrent administration of other vasodilators.

Individual variations: There is a large variation in hypotensive response between patients and this may be due to variable rate of absorption and first-pass metabolism during the passage of labetalol through the intestinal wall and liver.

Significant cardiomegaly

Clonidine: Concurrent use of beta-blockers and clonidine should be avoided because of the risk of adverse interaction and severe withdrawal symptoms. If administered concomitantly, the clonidine should not be discontinued until several days after the withdrawal of the beta-blocker.

Renal disease: Labetalol does not adversely affect renal function and is a particular suitable drug for use in hypertensive patients with renal disease. In patients with severe renal disease, haemodynamic changes following beta-blockade may impair renal function further.

Use in the Elderly: The bioavailability and half-life of Labetalol hydrochloride are increased in the elderly. In addition, the hypotensive response is greater in this age group following oral or i.v. administration. Therefore, lower doses of labetalol are likely to be required in elderly patients.

Use in Children: There is little reported clinical experience of the use of labetalol in children. Thus care should be taken in establishing individual dosage requirements in children. Safety and effectiveness in children have not been established.

INTERACTION WITH OTHER MEDICAMENTS:

Postural hypotension may be enhanced by the concurrent administration of other vasodilators. The interaction of labetalol with methyl dopa and with clonidine has been examined on blood pressure and heart rate in animals. The results indicate that labetalol, given together with methyl dopa or clonidine, should exert an additional hypotensive effect in human beings who are sensitive to both drugs in the combined therapy.

Concomitant use of tricyclic anti-depressants may increase the incidence of tremor.

Cimetidine increase the oral bioavailability of labetalol and care is required in determining the dose of labetalol in patients who are also taking cimetidine.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Beta-adrenergic blocking agents may cause bradycardia in the foetus and the newborn infant. During the final part of pregnancy and parturition these drugs should therefore only be given after weighing the needs of the mother against the risk to the foetus.

Labetalol is known to cross the placental barrier and has been found to bind to the eyes of foetal animals. Labetalol has been used successfully in the treatment of hypertension arising in the second and third trimester of pregnancy. Labetalol crosses the placental barrier and the possibility of the consequences of alpha- and beta- adrenoceptor blockade in the foetus and neonate should be borne in mind.

Perinatal and neonatal distress (bradycardia, hypotension, respiratory depression, hypoglycaemia, hypothermia) has been rarely reported.

Sometimes these symptoms developed a day or two after birth. Response to supportive measures (eg intravenous fluids and glucose) is usually prompt but with severe pre-eclampsia, particularly after prolonged intravenous labetalol, recovery may be slower. This may be related to diminished liver metabolism in premature babies. Intra-uterine and neonatal deaths have been reported but other drugs (eg vasodilators, respiratory depressants) and the effects of pre-eclampsia, intra-uterine growth retardation and prematurity were implicated. Such clinical experience warns against unduly prolonging high dose labetalol and delaying delivery and against co-administration of hydralazine. Administration of Labetalol in the first trimester of pregnancy is not recommended. Labetalol does not appear to be teratogenic in rats or rabbits, but it is embryolethal when given in a dose of 50 mg/kg orally.

Use in Lactation: Labetalol is excreted in breast milk. No adverse reactions in breast feeding infants have been reported. It is not recommended for nursing mothers unless the expected benefits outweigh any potential risks.

Nipple pain and Raynaud's phenomenon of the nipple have been reported.

SIDE EFFECTS:

Labetalol is normally well tolerated. Symptoms of postural hypotension may occur if the initial dosage is too high or if the dose is increased too rapidly.

These side effects include:

Cardiovascular: Postural hypotension may occur if the initial dosage is too high or if the dose is increased too rapidly. Occasionally bradycardia and heart block have been reported.

Central and Peripheral Nervous System: Transient dizziness, headache, tiredness, depressed mood and lethargy may occur. There have been reports of a tingling sensation of the skin (especially of the scalp) associated with Labetalol treatment usually occurring early in treatment and which is transient in nature.

Collagen disorders: There have been occasional reports of positive anti-nuclear antibodies unassociated with disease as well as the occasional case of systemic lupus erythematosus and very occasionally drug fever.

Eyes: Blurred vision, eye irritation and dry eyes have been reported.

Hepatic System: Reports of raised liver function tests, jaundice (both hepatocellular and cholestatic) and hepatitis have on occasion been reported the signs and symptoms of which are usually reversible on withdrawal of the drug. Hepatic necrosis has been reported (see Warnings).

Musculoskeletal System: There has been one report of toxic myopathy. Tremor has been reported in the treatment of hypertension of pregnancy. Muscle cramps have been reported.

Respiratory System: Bronchospasm may occur in susceptible individuals.

Skin and appendages: Rashes of various types such as generalised maculopapular, lichenoid, urticarial, bullous, lichen planus, psoriaform and facial erythema, Peyronies disease and reversible alopecia.

Urinary System: Acute retention of urine and difficulty in micturition have occurred during Labetalol treatment.

Gastrointestinal System: Epigastric pain, nausea and vomiting.

Hypersensitivity: Rash, pruritus, angioedema and dyspnoea.

Other Symptoms: Ankle swelling; failure of ejaculation; nasal congestion and sweating.

Reproductive System and Breast Disorders : Frequency 'not known': Nipple pain, Raynaud's phenomenon of the nipple

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage with labetalol causes excessive hypotension which is posture sensitive and sometimes, excessive bradycardia. Patients should be laid supine and, if necessary, their legs should be raised to improve the blood supply to the brain. Atropine 3 mg intravenously should be given to relieve the bradycardia. Gastric lavage or induced emesis is recommended if the overdosage is recent (2-3 hours). Oliguric renal failure has been reported after massive overdosage of labetalol orally.

If further measures are required to obtain adequate circulatory function, intravenous noradrenaline may be preferably to isoprenaline, the established pharmacological treatment for excessive cardiac beta-blockade. The recommended starting dose of noradrenaline in patients is 5-10 micrograms by intravenous injection, repeated as required, according to the response. Alternatively, noradrenaline may be infused at a rate of 5 micrograms / minute until a satisfactory response is achieved

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE: Please refer to outer package.

PACK SIZE: Blister pack of 25 x 10's.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.